Faculty of Science & Technology
PES's Modern College of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Level 7: First Year M. Tech. (2024 Pattern)

Course Code	CSE08651
Course Name	Skill Development – II
Semester	IV
Experiment No.	2
Title	Literature Mining using ChatGPT API

Problem Statement

To study and analyze the use of artificial intelligence-based tools for extracting, summarizing, and understanding information from large volumes of research literature.

Objective

To make use of the ChatGPT API for efficient literature mining, summarization, and knowledge extraction from scholarly research documents.

Outcome

CO653.2: Apply artificial intelligence tools for literature analysis and research support.

Software / Hardware Requirements

- ChatGPT API (OpenAI)
- Python Programming Environment
- Internet Connectivity
- Personal Computer or Laptop

Theory

Literature mining is an important research activity that involves extracting meaningful information, patterns, and insights from a large collection of scholarly documents. With the rapid growth of research publications, manually reviewing and analyzing literature has become time-consuming and inefficient. Therefore, automated tools based on artificial intelligence are increasingly used to support researchers in understanding and summarizing existing work.

Natural Language Processing (NLP) techniques play a crucial role in literature mining by enabling machines to understand, analyze, and generate human language. AI-based language models can process large textual datasets, identify key concepts, summarize content, and answer context-based queries. These capabilities help researchers quickly grasp the core contributions of research papers and identify relevant studies.

ChatGPT is an advanced AI language model that can be accessed through an application programming interface (API). It can be used for literature mining tasks such as paper summarization, keyword extraction, comparative analysis, and research question generation. By interacting with research text programmatically, the ChatGPT API supports efficient literature exploration and enhances research productivity.

Steps Involved

1. Access to the ChatGPT API was obtained through the OpenAI platform.
2. A Python environment was set up to interact with the API.
3. Research text or abstracts were provided as input to the ChatGPT model.
4. The API was used to generate summaries and extract key insights.
5. Relevant information was analyzed and documented for research use.

Results

The ChatGPT API was successfully used for literature mining. Research documents were summarized effectively, and key insights were extracted, enabling faster understanding of scholarly content.

Analysis and Conclusion

Analysis: The use of the ChatGPT API reduced the effort required for manual literature review and improved the efficiency of information extraction from research documents.

Conclusion: AI-based tools such as the ChatGPT API provide effective support for literature mining. Their application enhances research efficiency, enables faster knowledge acquisition, and assists researchers in managing large volumes of scholarly information.

References

1. OpenAI, "ChatGPT API Documentation," Available: https://platform.openai.com

2. Jurafsky, D., and Martin, J. H., *Speech and Language Processing*, Pearson.